20 Bias of Estimators

- MLE is one procedure for finding an estimator of a parameter θ. But when several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
  - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.

Example 20.1. Consider a large population in which 20% of individuals have an annual income ($ thousands) of 200, 40% of have an income of 70, and 40% have an income of 10. For this population, the population mean is

$$\mu = 10(0.4) + 70(0.4) + 200(0.2) = 72$$

Also, the population variance is

$$[10^2(0.4) + 70^2(0.4) + 200^2(0.2)] - 72^2 = 4816$$

and the the population SD is $\sigma = 69.4$.

Suppose our goal is to estimate σ based on a sample of size n. We'll start by estimating the population variance σ^2. In this example, we know the population distribution and we know the population variance $\sigma^2 = 4816$, but in practice the population distribution and the population variance would be unknown.

One estimator of the population variance σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

1. For $n = 2$, describe in detail how you would use simulation to approximate the distribution of $\hat{\sigma}^2$, and its expected value.

2. For $n = 2$ determine the distribution of $\hat{\sigma}^2$. (This is not a simulation; use ideas from the first half of the course.)

3. Find and interpret $\mathrm{P}(\hat{\sigma}^2 = 0)$.

4. Compute and interpret $\mathrm{E}(\hat{\sigma}^2)$. Compare $\mathrm{E}(\hat{\sigma}^2)$ to σ^2; what does this say?

- $\hat{\theta}$ is an **unbiased** estimator of θ if

$$\mathrm{E}\left(\hat{\theta}\right) = \theta \quad \text{for } \textit{all} \text{ potential values of } \theta$$

- Roughly speaking, an unbiased estimator does not tend to systematically overestimate or underestimate the parameter of interest, regardless of the true value of the parameter.
- The **bias (function)** of an estimator is the difference between its expected value (over many hypothetical samples) and the true (but unknown) value of the parameter

$$\text{bias}_\theta(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta) = \mathrm{E}(\hat{\theta}) - \theta, \qquad \text{a function of } \theta$$

Example 20.2. Continuing Example 20.1. Another commonly used definition of the sample variance is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

1. For $n = 2$, determine the distribution of S^2.

2. Compute and interpret $\mathrm{E}(S^2)$. Compare $\mathrm{E}(S^2)$ to σ^2; what does this say?

3. The *sample standard deviation* is commonly defined as $S = \sqrt{S^2}$, that is,

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}$$

   For $n = 2$, determine the distribution of S.

4. Compute and interpret $\mathrm{E}(S)$. Compare $\mathrm{E}(S)$ to σ; what does this say?

- For a random sample $X_1, \ldots, X_n$, the **sample variance** is commonly defined as

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

- The **sample standard deviation** is $S = \sqrt{S^2}$.

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}$$

- The sample variance S^2 is an *unbiased* estimator of the population variance σ^2

$$\mathrm{E}(S^2) = \sigma^2$$

- However, the sample SD S is a *biased* estimator of the population SD σ

$$\mathrm{E}(S) < \sigma$$

  - Though the bias decreases as the sample size increases

Example 20.3. Let $X_1, \dots, X_n$ be an (i.i.d.) random sample from a population with population mean μ. Let $\bar{X} = (X_1 + \dots + X_n)/n$ be the sample mean. It can be shown that

$$\mathrm{E}(\bar{X}) = \mu$$

There are three means above; explain what all the means mean, and what the equation says.

- The **sample mean** is

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

- The sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ.

$$\mathrm{E}(\bar{X}_n) = \mu$$

- Over many random samples, sample means do not systematically overestimate or underestimate the population mean.

Example 20.4. Recall that in the car dealership problem we were trying to estimate μ for a Poisson(μ) distribution. We considered a few different estimators of μ. Determine which of the following estimators are unbiased estimators of μ based on a random sample $X_1, \dots, X_n$ of size n from a Poisson(μ) distribution. If the estimator is not unbiased, identify its bias function.

1. $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

2. $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$

3. $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$. (Hint: $\hat{\sigma}^2 = \frac{n-1}{n}S^2$.)

4. $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$. (Recall this estimator was like a weighted average of the mean from the old dealership and the sample mean for the new dealership.)

5. 2.3. (This estimator ignores the sample data from the new dealership entirely and just uses the mean from the old dealsership as the estimator.)

Example 20.5. Continuing the Poisson(μ) problem. Suppose $n = 3$. Imagine we had not already derived the bias function of the estimator $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$

1. Describe in detail how you would use simulation to approximate the bias of $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$ when $\mu = 2.3$.

2. Describe in detail how you would use simulation to approximate the bias *function* of $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$.

Example 20.6. Continuing the Poisson(μ) problem. Let $\theta = e^{-\mu}$. We know $\bar{X}$ is an unbiased estimator of μ. We also know that $e^{-\bar{X}}$ is the MLE of $e^{-\mu}$. Is $\hat{\theta} = e^{-\bar{X}}$ an unbiased estimator of $\theta = e^{-\mu}$?

- In general, if $\hat{\theta}$ is an unbiased estimator of θ, $g(\hat{\theta})$ is usually NOT an unbiased estimator of $g(\theta)$ (unless g is linear).
  - Because in general $\mathrm{E}(g(\hat{\theta})) \neq g(\mathrm{E}(\hat{\theta}))$.

Example 20.7. Continuing the Poisson(μ) problem. Consider the estimator $\hat{\sigma}^2$. Recall that the bias function is $\text{bias}_\mu(\hat{\sigma}^2) = -\frac{\mu}{n}$. What happens to the bias function as $n \to \infty$. What does this mean?

- An estimator $\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if $\mathrm{E}(\hat{\theta}_n) \to \theta$ as $n \to \infty$ for all θ, that is, if $\text{bias}_\theta(\hat{\theta}_n) \to 0$ as $n \to \infty$ for all θ.
- For an asymptotically unbiased estimator, if the sample size is large $\mathrm{E}(\hat{\theta}) \approx \theta$, regardless of the true value of θ. That is, if the sample size is large the estimator is "nearly unbiased".
- MLEs are often biased, but in general, MLEs are asymptotically unbiased

21 Variance of Estimators

- The bias function of an estimator measures the estimator's tendency to overestimate or underestimate the parameter, as a function of potential values of the parameter.
- But bias is only one consideration. Bias only considers the average value of the estimator over many samples, which is just one feature of the estimator's sampling distribution.

Example 21.1. Consider a large population in which 20% of individuals have an annual income ($ thousands) of 200, 40% of have an income of 70, and 40% have an income of 10. For this population, the population mean is

$$\mu = 10(0.4) + 70(0.4) + 200(0.2) = 72$$

Also, the population variance is

$$[10^2(0.4) + 70^2(0.4) + 200^2(0.2)] - 72^2 = 4816$$

and the the population SD is $\sigma = 69.4$.

Suppose our goal is to estimate μ based on a sample of size n. In this example, we know the population distribution and we know the population mean $\mu = 72$, but in practice the population distribution and the population mean would be unknown.

One estimator of the population mean μ is

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

First suppose $n = 2$.

1. Determine the distribution of $\bar{X}$.

2. Find and interpret $\mathrm{P}(\bar{X} \geq 105)$.

3. Compute $\mathrm{E}(\bar{X})$. How does it relate to μ? Why?

4. Compute and interpret $\mathrm{Var}(\bar{X})$. How does it relate to the population variance σ^2?

5. Compute and interpret $\mathrm{SD}(\bar{X})$. How does it relate to the population standard deviation σ?

- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)
- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
  - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
  - The probability distribution of a statistic is called a **sampling distribution**.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* and computing the value of the statistic for each sample.
  - The resulting standard deviation, called the **standard error**, measures the sample-to-sample variability of the statistic over many (hypothetical) samples of the same size.

- However, *in practice usually only a single sample is selected* and a single value of the statistic is observed.

- For many statistics (e.g., means and proportions) — but not all — statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.

Example 21.2. Continuing Example 21.1, now consider $n = 8$.

1. How could you use simulation to approximate the distribution of $\bar{X}$, and its expected value and SD?

2. Find and interpret $\mathrm{E}(\bar{X})$ without first finding the distribution of $\bar{X}$, and compare to the simulation results.

3. Find and interpret $\mathrm{SD}(\bar{X})$ without first finding the distribution of $\bar{X}$, and compare to the simulation results. What does this measure variability of? Compare to $n = 2$.

4. Use simulation to find and interpret $\mathrm{P}(\bar{X} \ge 105)$. Compare to $n = 2$.

- For an i.i.d. random sample $X_1, \dots, X_n$, the **sample mean** is the *statistic*

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{X_1 + \cdots + X_n}{n}$$

- The sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ:

$$\mathrm{E}(\bar{X}_n) = \mu$$

  - Over many random samples, sample means do not consistently overestimate or consistently underestimate the population mean

- Sample-to-sample variability of sample means decreases as the sample size increases

$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

$$\text{sample-to-sample SD of sample means} = \frac{\text{unit-to-unit SD of values of the variable}}{\sqrt{\text{sample size}}}$$

  - The more variable the individual values of the variable are (i.e. the larger the population SD of the variable, σ), the more variable sample means will be (i.e. the larger than the SE).
  - Sample means are less variable than individual values of the variable
  - Over many random samples, sample means from larger random samples vary less, from sample to sample, than sample means from smaller random samples.
  - The SD of the sample means decreases as the sample size increases, but this reduction in SD follows a "square root rule". For example, if the sample size is increased by a factor of 4, then the SD is reduced by a factor of 2 (not 4).

Example 21.3. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Assume $n = 3$ and $\mu = 2.3$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of $\bar{X}$ and its and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

2. Assume $n = 3$ and $\mu = 2.3$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of S^2 and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

3. Compare the simulation results for $\bar{X}$ and S^2 when $n = 3$ and $\mu = 2.3$. Based on the simulation results, which estimator of μ is preferred when $\mu = 2.3$ — $\bar{X}$ or S^2? Why? But then explain why this information isn't very helpful.

4. Assuming $n = 3$ and $\mu = 2.3$, compute $\mathrm{E}(\bar{X})$ and $\mathrm{SD}(\bar{X})$, and compare to the simulation results.

5. For a general n and $\mu > 0$, find expressions for $\mathrm{E}(\bar{X})$ and $\mathrm{SD}(\bar{X})$.

6. It can be shown that
$$\mathrm{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}$$
Sketch a plot of the variance functions of both $\bar{X}$ and S^2 as a function of μ. Regardless of the value of μ, which estimator has smaller variance? Which of these two unbiased estimators of μ, $\bar{X}$ or S^2, is preferred?

- When choosing between two *unbiased* estimators, the one with smaller variance is generally preferred.
- Remember that the variance of an estimator will be a function of the unknown parameter θ, so we need to consider the variance *function* for various potential values of θ.

22 Mean Square Error of an Estimator

- MLE is one procedure for finding an estimator of a parameter θ. But when several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
  - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.

Example 22.1. Continuing the situation of estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. Consider the estimators

$$S^2 = \frac{1}{n-1}\sum_{i=1}^n \left(X_i - \bar{X}\right)^2$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \left(X_i - \bar{X}\right)^2 = \left(\frac{n-1}{n}\right) S^2$$

We have seen that S^2 is an unbiased estimator of μ but $\hat{\sigma}^2$ is not. Does that mean we should prefer S^2 over $\hat{\sigma}^2$?

1. Assume $n = 3$ and $\mu = 2.3$. Use simulation to approximate the distribution of $\hat{\sigma}^2$ and its expected value and standard deviation.

2. Compare the simulation results to those for S^2 (from a previous handout.) For which estimator is the bias smaller? For which estimator is the variance smaller? Is there a clear preference between the two estimators when $\mu = 2.3$?

- Estimators can be evaluated based on both bias and variability.
- Mean square error is a combined measure of both an estimator's bias and its variability.
- The **mean square error (MSE)** of an estimator $\hat{\theta}$ of a parameter θ is

$$\text{MSE}_\theta(\hat{\theta}) = \text{E}\left(\left(\hat{\theta} - \theta\right)^2\right), \qquad \text{a function of } \theta$$

- Mean square error measures, on average, how far the estimator deviates from the parameter it is estimating.
- The MSE of an estimator is a function of the parameter θ. It can be shown that[1]

$$\begin{aligned}\text{MSE}_\theta(\hat{\theta}) &= \text{Var}(\hat{\theta}) + \left(\text{E}(\hat{\theta}) - \theta\right)^2 \\ &= \text{Var}(\hat{\theta}) + \left(\text{bias}_\theta(\hat{\theta})\right)^2\end{aligned}$$

- There can be many "reasonable" estimators of a parameter. Given two estimators, it's often the situation that neither one has smaller MSE for all potential values of θ. Choosing an estimator often involves a tradeoff between bias and variability.

3. Use the simulation results to compute the MSE of S^2 and $\hat{\sigma}^2$ when $\mu = 2.3$. Determine which estimator has smaller MSE when $\mu = 2.3$, but then explain why this information is not very useful.

[1]Recall: By definition $\text{Var}(Y) = \text{E}[(Y - \text{E}(Y))^2]$ but remember the useful computational formula $\text{Var}(Y) = \text{E}(Y^2) - (\text{E}(Y))^2$. Apply this to $Y = \hat{\theta} - \theta$, and remember that $\hat{\theta}$ is random but θ is not.

4. Recall
$$\mathrm{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}$$
Find and plot the MSE functions of S^2 and $\hat{\sigma}^2$. Which estimator is preferred? Discuss.

Example 22.2. Continuing estimating μ for a Poisson(μ) distribution. Consider $\bar{X}$ and the constant estimator 2.3.

1. Find the MSE of the constant estimator 2.3.

2. Find the MSE of $\bar{X}$.

3. Suppose $n = 3$. Plot the MSE functions of the two estimators. Does either estimator have a better MSE? Explain.

4. Donny Don't says: "neither estimator has smaller MSE for all values of μ. So we'll use the constant estimator 2.3 when μ is near 2.3 (between 1.57 and 3.36 if $n = 3$) and we'll use $\bar{X}$ for other values of μ". Do you agree with Donny's strategy? Do you see any problems with it?

5. What happens to the MSEs as n increases? In particular, what happens to the range of values of μ for which the constant estimator 2.3 has smaller MSE than $\bar{X}$?

- Given a parameter θ, it is never possible to find a single estimator that has the smallest MSE for all potential values of θ.
- Choosing an estimator often involves a tradeoff between bias and variability.
- MLEs generally have good properties. For many situations, roughly,
  - The MLE is asymptotically unbiased; that is, the bias is small when the sample size is large
  - The variance of an MLE is about as small as it can be (for an unbiased estimator)
- But there are many reasonable estimation procedures besides MLE (e.g., Bayes estimators).
- Bias is estimation is not necessarily bad. Being willing to accept a little bit of bias can often lead to a beneficial reduction in variability.
  - On the other hand, try to minimize bias in *data collection*: how the sample is selected, how the variables are measured, etc
  - A lot of the theory assumes we have a perfect random sample, which is never true in practice